8/1/26

NLP

Dr. Swapnil Hingmire

→ Language processing in natural language.
→ Processing of natural language

→ Computational study of language use
→ Poetry : meaning conveyed not explicitely
⇒ difficult to process.
→ Meaning: acquire, represent, retrieve

Marr's Levels of Analysis

→ Computational
- goal? - i/p? - o/p? - math
→ Algorithmic
- what representations & algorithms
→ Implementation
- how? - physically? - software?

eg: Modelling people waiting in a line
computational: FIFO queue
algorithm: ~~Stact~~ Linked list data structure
implementation: pointers, memory management

Meaning: Summary / Translation / Sentiment / QA / Generate

→ Word by word / Sentence by sentence translation of poetry may not work.
Context, rhyme scheme, semantic meaning is lost

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→ Hypo cognition: Things that don't have words (Some emotions don't have words ⇒ people don't know what is wrong)

→ Hyper cognition: eg: depression (lunch was bad. I'm depressed. Such sentences have reduced the meaning of depression. Now ppl take it casually)

APPLICATIONS

→ Machine Translation
→ Speech Recognition
→ Search Engines
⋮

TASK

→ Part of speech tagging
→ Dependency parsing
→ Named Entity Recognition
⋮
→ Coreference resolution
eg: a and b are friend. He is helping her
refers to "b"
refers to a

Why NLP is Hard?

→ Optimization
Conserve energy
But convey maximum meaning
→ Space & Time
→ Cognition
→ Frames: Schema
→ Subjectivity (people have different opinions) Politics, Religion, culture, poetry, ...
→ Context & Intent
eg: Change will not given. You have to bring change
If bus conductor says this, meaning is very different from when a politician says this
→ The world

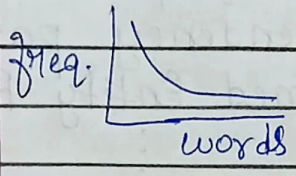
Moravec's Paradox
"Easy things for us are difficult for computers."
A lot of meaning is conveyed implicitely.

Gestalt Principles: (i) Emergence
(ii) Invariance
(iii) Continuity

→ Though NLP is hard, we have some known

patterns or conventions.

→ Power law.



Few words are very frequent
Remaining (many) words are less frequent.

⇒ TEXT OR UTTERANCE IS NOT DATA

Common misconceptions

→ Language is not about "words" & what they mean

→ It is about "people" & what they mean